

# DINH NHAT LUAN

## AI Engineer Intern

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## PROFESSIONAL SUMMARY

Information Systems student at UIT with hands-on experience building a Vietnamese legal search engine and data-driven web applications. Developed **Python pipelines** for corpus processing, **hybrid retrieval**, **evaluation**, and **REST APIs** using **FastAPI**, **pandas**, **NumPy**, **scikit-learn**, **Docker**, and **Git**. Comfortable working with **SQL**, structured data, **JSON artifacts**, **model evaluation**, and **reproducible testing**. Interested in applying **AI**, retrieval, and automation to **Marketing & Product workflows** while expanding practical experience with **LLM APIs**, **prompt engineering**, and **tool-calling agents**.

## EDUCATION

VNU-HCM University of Information Technology (UIT)

2023 - 2026

Major: Information Systems

Relevant Coursework: Database Systems, Web Application Development, OOP, Software Engineering

## KEY PROJECTS

**Vietnamese Legal Semantic Search Engine | Python, FastAPI, TF-IDF, E5, RRF, Cross-Encoder, Docker |**

GitHub: <https://github.com/dustin-dinh/vietnamese-legal-semantic-search>

- **Problem:** Workers often describe labor-law issues in **everyday Vietnamese**, **without accents** or with **abbreviations**, while **exact keyword search** struggles to map those questions to formal legal terminology.
- **Applied solution:** Built a **retrieval-only pipeline** over **220 articles** and **228 chunks**, combining **accented/accentsless TF-IDF**, **multilingual E5 embeddings**, **weighted RRF**, and **Cross-Encoder reranking** to return original provisions with **article**, **PDF-page**, and **source metadata**.
- **Evaluation:** Achieved **Hit@5 of 0.90** and **MRR@5 of 0.7244** on a **locked 80-query test set**; added **confidence thresholds**, **clarification**, and **out-of-scope abstention** to avoid low-relevance results.
- **Engineering reliability:** Exposed retrieval through **FastAPI** and **Streamlit**, verified **30/30 versioned artifacts**, passed a **10/10 corpus audit** and **33 offline pytest cases**, and documented alpha limitations and data provenance.

**Furniture Shop Web Application | React 19, Laravel 12, MySQL, REST API, Stripe |**

GitHub: <https://github.com/dustin-dinh/furniture-shop-web-application>

- **Problem:** **Product**, **cart**, **customer**, **payment**, and **order data** can become **inconsistent** when storefront and admin workflows are built without **clear system and data boundaries**.
- **Applied solution:** Built a decoupled **React SPA** and **Laravel REST API** with **normalized MySQL entities**, **Sanctum authentication**, separated customer/admin workflows, **filtered and paginated catalog**, **persistent cart**, **COD/Stripe checkout**, and **order history**.
- **Workflow automation:** Implemented **admin flows** for products, categories, users, orders, shipping charges, **dashboard reporting**, and **temporary image upload** with **large/small image processing**.
- **Project scale:** Structured **14 controllers**, **9 Eloquent models**, **14 migrations**, **56 JSX files**, and **roughly 40 API route declarations**; all **64 PHP files** passed syntax checks.

## TECHNICAL SKILLS

**AI & Retrieval:** Hybrid retrieval, semantic search, TF-IDF, dense embeddings, vector search, query normalization, reranking, confidence filtering, **Hit@K/MRR evaluation**, retrieval/RAG foundations.

**Programming & Data:** Python, pandas, NumPy, scikit-learn, SQL, JavaScript, JSON/CSV processing, data validation, REST API integration, basic ETL and data pipelines.

**Backend & Databases:** FastAPI, Streamlit, PostgreSQL/pgvector, MySQL, Laravel REST API, relational database design, normalization.

**Tools & Engineering:** Git, GitHub, Docker, Pytest, Ruff, Vite, Linux/CLI basics, artifact versioning and reproducible testing.

## WORK STYLE & LANGUAGES

**Work Style:** Independent project ownership; effective use of **AI tools** for research, implementation support, debugging, and documentation.

**Languages:** Vietnamese: Native | English: TOEIC 700 / B2 equivalent.